Data Mining Applications

A Generic Model for End State Prediction of Business Processes Towards Target Compliance

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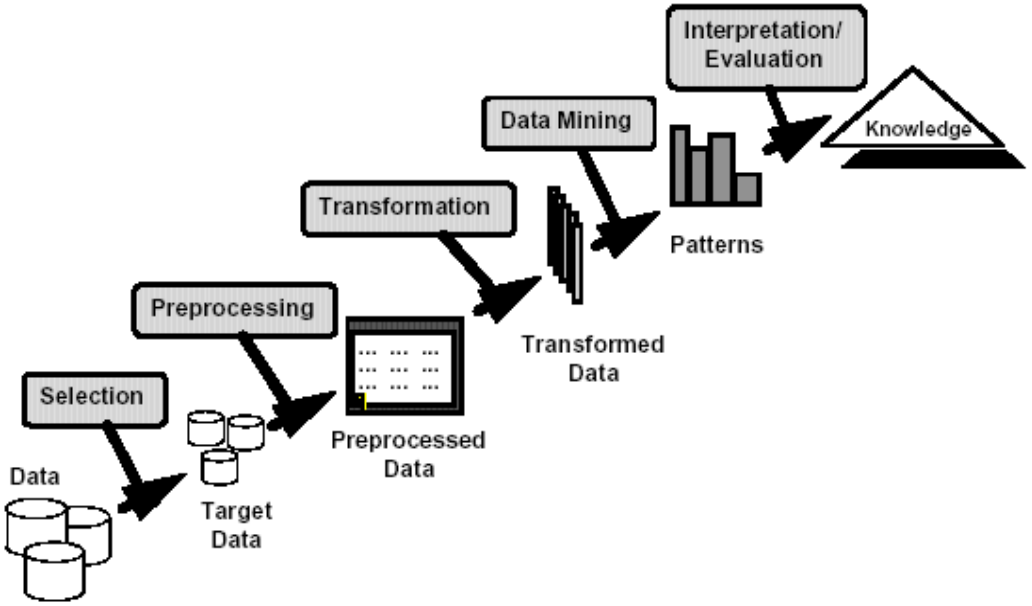
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Data mining

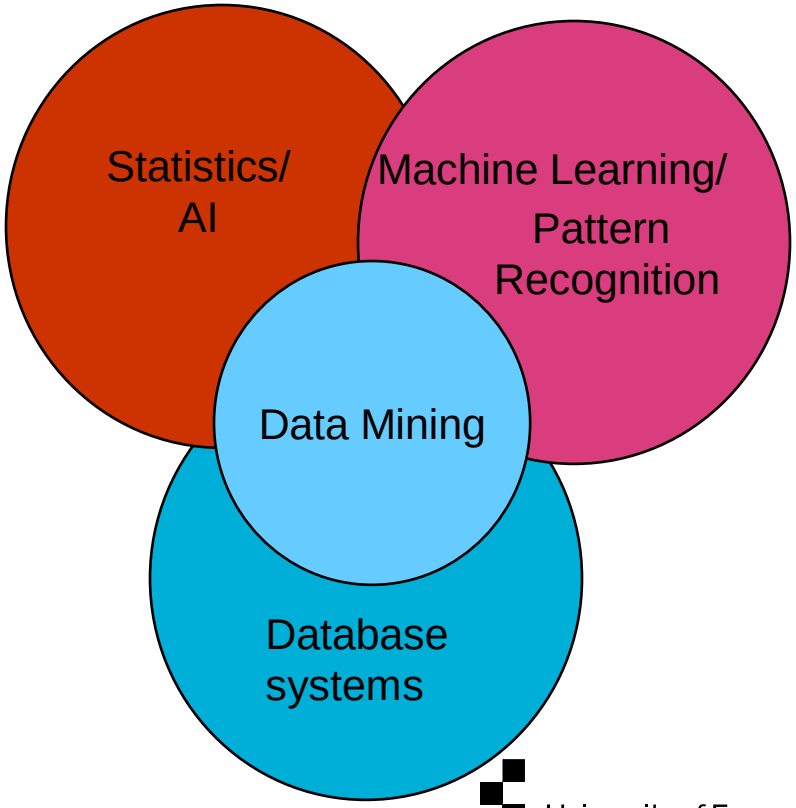
Many Definitions

- Non-trivial extraction of implicit, previously unknown and potentially useful information from data
- Exploration & analysis, by automatic or semi-automatic means, of large quantities of data in order to discover meaningful patterns



Origins of Data Mining

- Draws ideas from machine learning/AI, pattern recognition, statistics, and database systems
- Traditional Techniques may be unsuitable due to
  - Enormity of data
  - High dimensionality of data
  - Heterogeneous, distributed nature of data



Problem Statement

- When a customer placed an order, a process with a unique ID is generated.
- The generated process will be executed through a series of tasks to complete the order.
- The ultimate purpose of the data is to develop a system for early prediction of an order, i.e. the order will be delayed/cancelled or delivered on-time.

Classification VS Prediction

- Determining the category of an order after its completion is referred as classification problem.
- Determining the category of an order before its completion is referred as prediction problem.
- Prediction problems are difficult as compared to classification problems.
- Our aim is prediction of an order, i.e. it is going to be delivered on-time or not.

Relational Schemas in ORDER Database

Process(*ProcessID*, orderDateTime, comitmentDateTime, Department, ProductID, …….)

Task(ProcessID, taskID, startDateTime, endDateTime, ……..)

Problems with the data:

- **endDateTime** is not recorded
- When tasks of a process are extracted, they are not in the order of occurrence

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- Total number of processes in extracted dataset is 500,000
- Approximately 1200 different tasks exist in the dataset
- For example, 32 tasks are associate with process 551485549
- Tasks associated with a process vary in number as well as description.

Task associated with the process 551485549.

Sr. No.	Task ID	Description	Starting Time
1	100188		'22/04/2018 11:36:07'
2	100197		'03/04/2018 16:34:50'
3	100198		'11/04/2018 13:39:49'
4	100198		'11/04/2018 13:56:36'
5	100199		'03/04/2018 14:58:01'
6	100350		'11/04/2018 07:55:11'
7	100372		'03/04/2018 14:55:20'
8	100442		'03/04/2018 14:41:19'
9	100442		'06/07/2018 15:49:28'
10	100443		'03/04/2018 15:02:18'
11	100465		'11/04/2018 11:17:35'
12	100500		'03/04/2018 14:58:20'
13	100575		'03/04/2018 16:27:15'
14	100689		'04/04/2018 15:21:26'
15	100689		'10/04/2018 02:57:53'
16	100748		'11/04/2018 11:24:36'
17	100887		'11/04/2018 13:41:53'
18	100887		'11/04/2018 13:54:21'
19	100921		'11/04/2018 13:41:53'
20	100958		'11/04/2018 07:55:11'
21	100979		'03/04/2018 15:32:41'
22	100992		'08/04/2018 02:49:25'
23	100992		'08/04/2018 02:49:26'
24	101045		'12/04/2018 01:03:29'
25	101046		'10/04/2018 02:57:53'
26	83966380		'11/04/2018 11:17:05'
27	433197388		'11/04/2018 13:56:36'
28	433197388		'12/04/2018 10:10:42'
29	433447454		'12/04/2018 10:10:42'
30	508370098		'19/04/2018 21:17:45'
31	508370098		'23/04/2018 09:56:41'
32	508370098		'06/06/2018 12:47:25'

- As ending time of the tasks are missing, therefore, they are sorted in order of occurrence to determine their completion time

Sr. No.	Task ID	Description	Sorted Starting Time
1	100442	'Order Created'	'03/04/2018 14:41:19'
2	100372	'Order Submitted'	'03/04/2018 14:55:20'
3	100199		'03/04/2018 14:58:01'
4	100500		'03/04/2018 14:58:20'
5	100443		'03/04/2018 15:02:18'
6	100979		'03/04/2018 15:32:41'
7	100575		'03/04/2018 16:27:15'
8	100197		'03/04/2018 16:34:50'
9	100689		'04/04/2018 15:21:26'
10	100992		'08/04/2018 02:49:25'
11	100992		'08/04/2018 02:49:26'
12	100689		'10/04/2018 02:57:53'
13	101046		'10/04/2018 02:57:53'
14	100958		'11/04/2018 07:55:11'
15	100350		'11/04/2018 07:55:11'
16	83966380		'11/04/2018 11:17:05'
17	100465		'11/04/2018 11:17:35'
18	100748		'11/04/2018 11:24:36'
19	100198		'11/04/2018 13:39:49'
20	100887		'11/04/2018 13:41:53'
21	100921		'11/04/2018 13:41:53'
22	100887		'11/04/2018 13:54:21'
23	100198		'11/04/2018 13:56:36'
24	433197388		'11/04/2018 13:56:36'
25	101045		'12/04/2018 01:03:29'
26	433447454		'12/04/2018 10:10:42'
27	433197388		'12/04/2018 10:10:42'
28	508370098		'19/04/2018 21:17:45'
29	100188		'22/04/2018 11:36:07'
30	508370098		'23/04/2018 09:56:41'
31	508370098		'06/06/2018 12:47:25'
32	100442		'06/07/2018 15:49:28'

- Time of completion of all tasks associated with the process 551485549
- The completion time for the last task cannot be computed due to unavailability of its ending time and the starting time of the next task.

Sr. No.	Task ID	Description	Time of Completion of a Task T_i
1	100442	'Order Created'	'00:14:01'
2	100372	'Order Submitted'	'00:02:41'
3	100199		'00:00:19'
4	100500		'00:03:58'
5	100443		'00:30:23'
6	100979		'00:54:34'
7	100575		'00:07:35'
8	100197		'22:46:36'
9	100689		'83:27:59'
10	100992		'00:00:01'
11	100992		'48:08:27'
12	100689		'00:00:00'
13	101046		'28:57:18'
14	100958		'00:00:00'
15	100350		'03:21:54'
16	83966380		'00:00:30'
17	100465		'00:07:01'
18	100748		'02:15:13'
19	100198		'00:02:04'
20	100887		'00:00:00'
21	100921		'00:12:28'
22	100887		'00:02:15'
23	100198		'00:00:00'
24	433197388		'11:06:53'
25	101045		'09:07:13'
26	433447454		'00:00:00'
27	433197388		'179:07:03'
28	508370098		'62:18:22'
29	100188		'22:20:34'
30	508370098		'1058:50:44'
31	508370098		'723:02:03'
32	100442		Not available

- The appearance of tasks in a process with or without repetition can also be a crucial feature for prediction.
- Therefore, unique and repeated tasks (with number of repetition) for each process are computed
- **19** out of 32 tasks are appeared only once whereas **6** tasks are repeated.

Sr. No.	ID of Unique Tasks	ID of Repeated Tasks	Number of times a task Repeated n_r
1	100442	100198	2
2	100372	100689	2
3	100199	100887	2
4	100500	100992	2
5	100443	433197388	2
6	100979	508370098	3
7	100575		
8	100197		
9	101046		
10	100958		
11	100350		
12	83966380		
13	100465		
14	100748		
15	100921		
16	101045		
17	433447454		
18	100188		
19	100442		

Unique and repeated tasks (with number of repetition)

ID of Repeated Tasks	Number of times a task Repeated n_r	Total time for each Repeated Task
100198	2	'00:02:04'
100689	2	'83:27:59'
100887	2	'00:02:15'
100992	2	'48:08:28'
433197388	2	'190:13:56'
508370098	3	'1844:11:09'

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Extracted Features

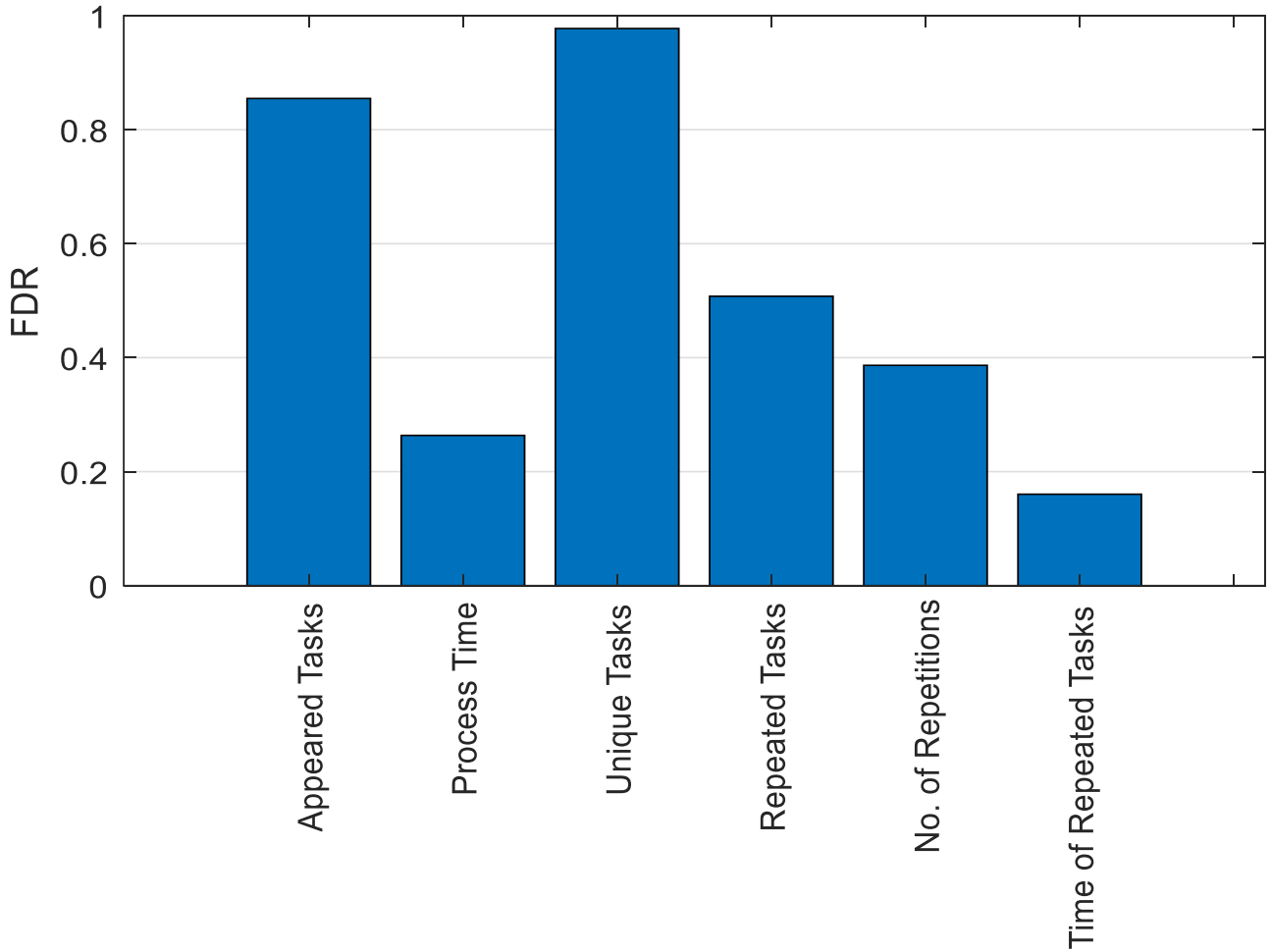
- Appeared Tasks
- Process Time
- Unique Tasks
- Repeated Tasks
- Number of Repetitions
- Time of Repeated Tasks

For the process 551485549, this feature looks like

F = [32 2257.1 19 6 13 2166.1]

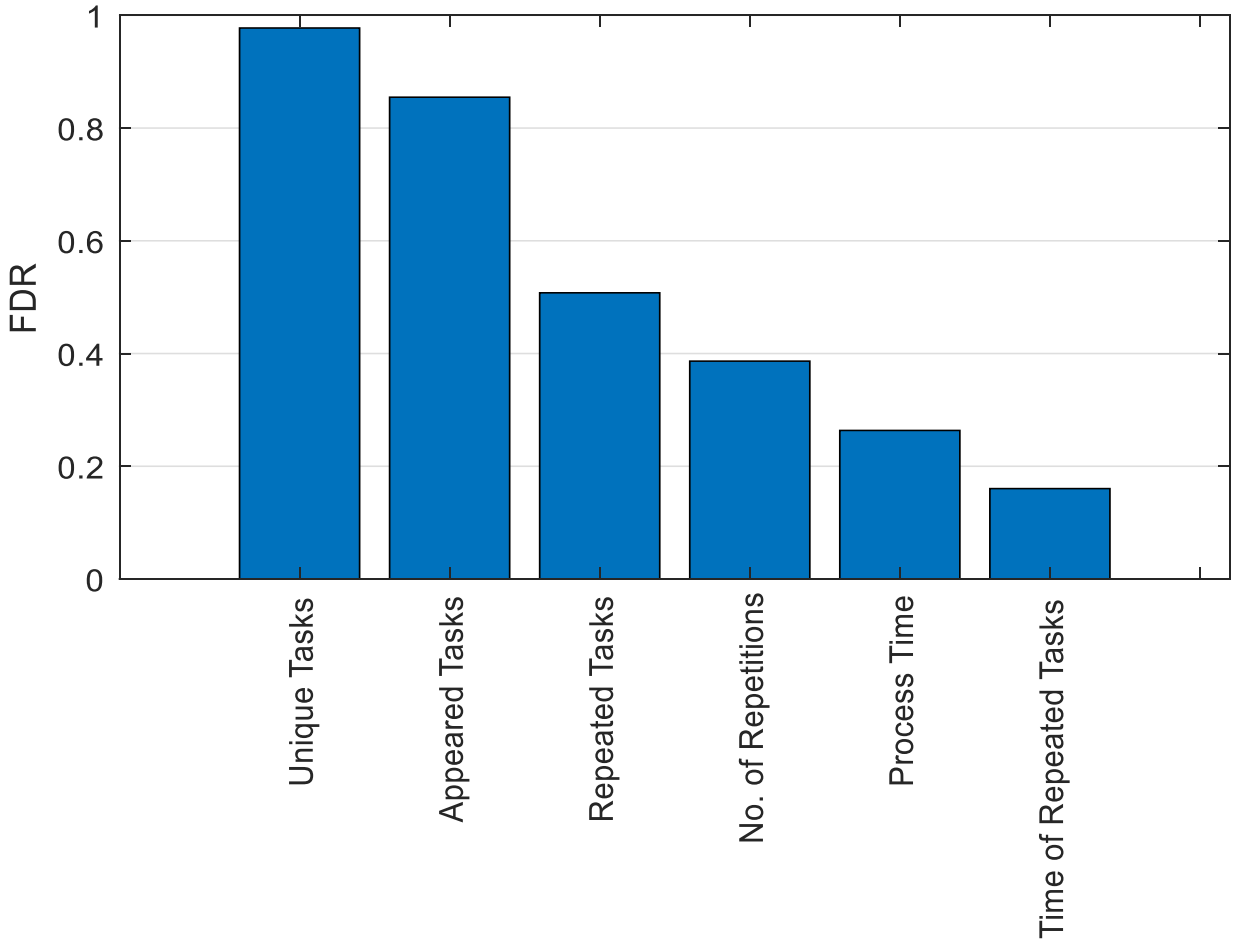
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Fisher Discriminant Ratio of Each Feature (1)



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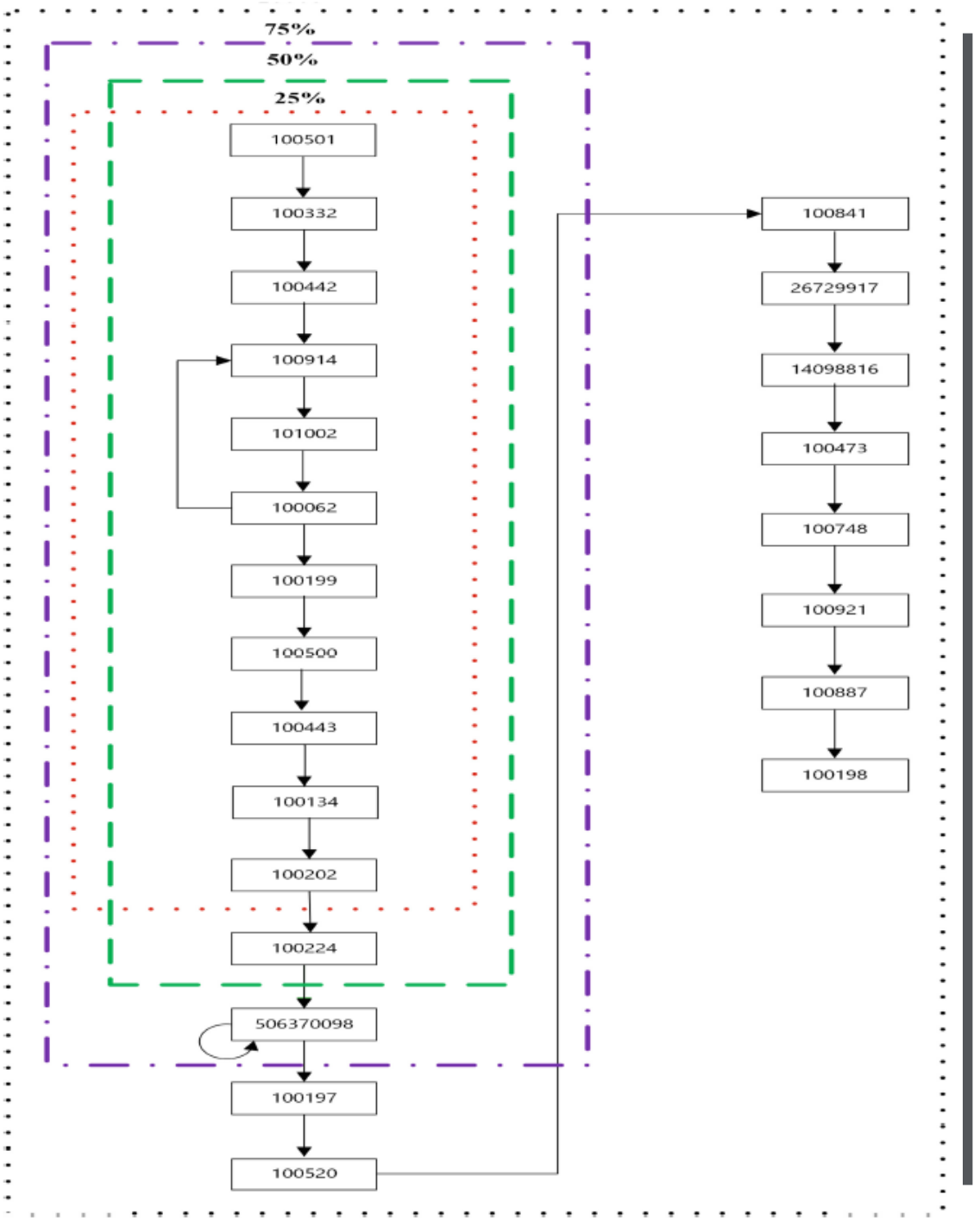
Fisher Discriminant Ratio of Each Feature in Descending Order



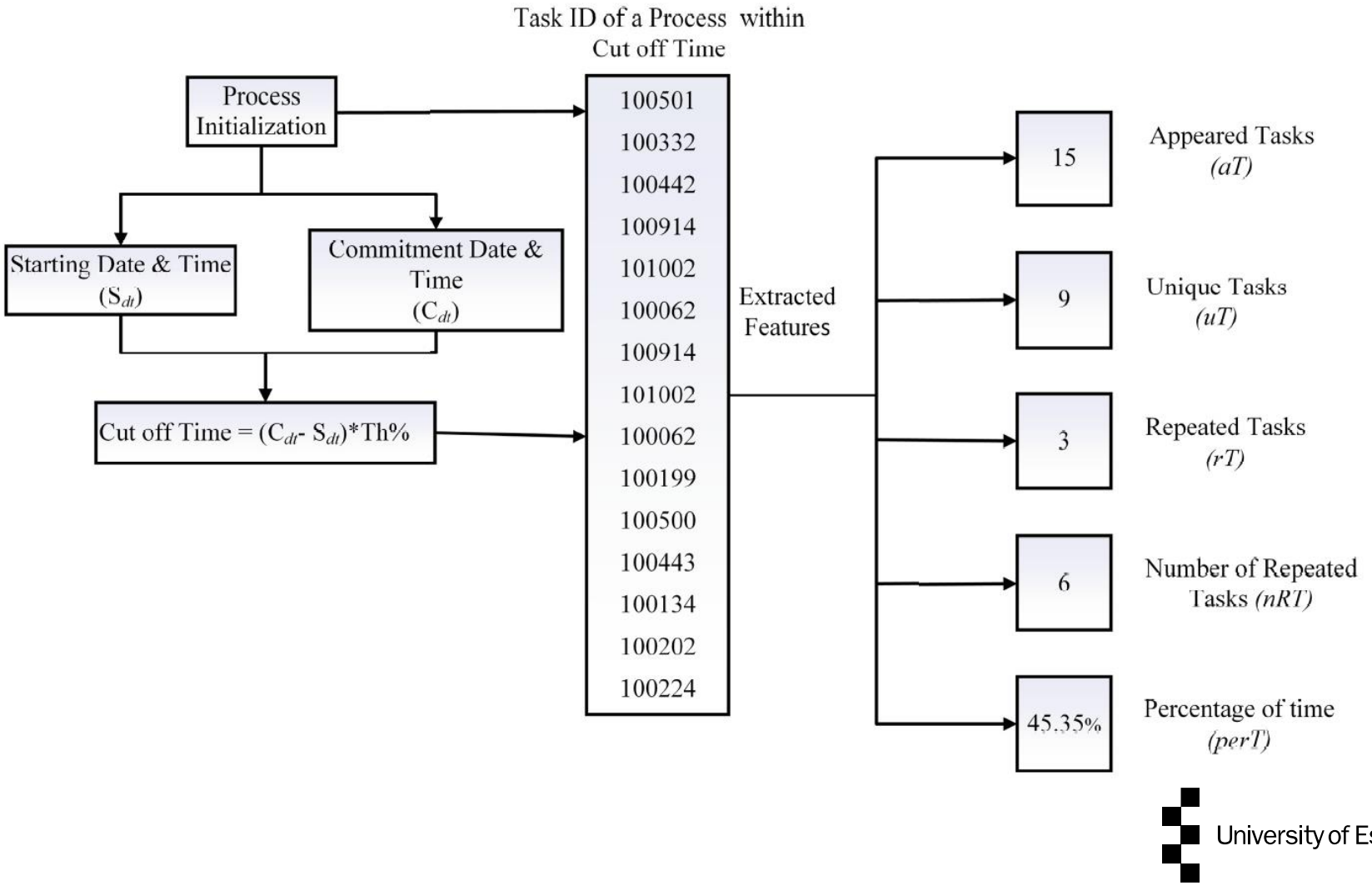
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- T_{dt} is the Target date and time
- S_{dt} is the process starting date and time
- *Th*% is the 25%, 50%, 75%, 85%, 95% and 100% is a threshold.

$$Cut - off\ \ Time = (T_{dt} - S_{dt}) * Th\%$$

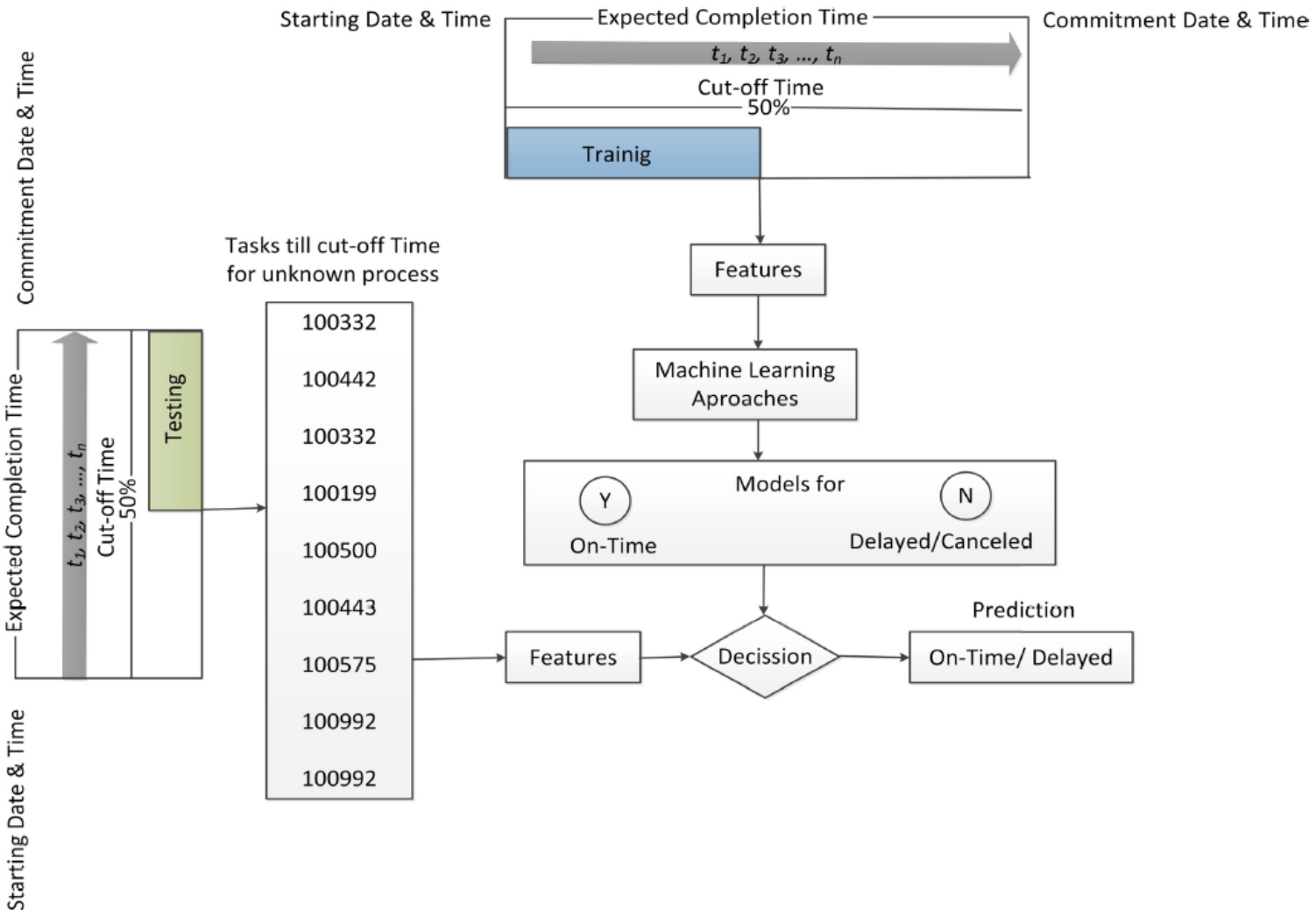


Feature Extraction Process for Prediction of Order



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Prediction Process



Result at 50% Cut-off Time

Algorithm	50% cut-off time					
	Accuracy %	Precision %	Sensitivity %	Specificity %	F-measure %	AUC
Logistic Regression	75.10	92.22	77.03	65.33	83.85	0.74
J48	75.07	84.95	80.59	59.36	82.71	0.68
SVM	75.07	90.45	77.70	63.32	83.59	0.74
Naive Bayes	55.72	44.14	85.95	38.67	58.33	0.72

Conclusion

- The monitoring of a business processes to avoid delay in the delivery of a customer's order is a crucial element for better customer experience and to avoid financial loss for the company.
- The proposed system tries to provide a solution to this problem, where the real issue is to provide a means of monitoring the progress of an order in real time and assess whether we can infer a likely breach of target compliance and predict delay at an early stage.
- Finding the optimal time for intervention and mediating against the evolving situation is a trade-off between having confidence in our prediction and intervening in a timely manner.